

Analytic spaces

Fix a complete non-archimedean field $\mathbf{k}$.

1 Definition of analytic spaces

Definition 1. Let X be a topological space. A *net* (resp. *quasi-net*) on X is a collection τ of subsets of X such that

- (a) for every $x \in X$, there exist $U_1, \dots, U_n \in \tau$ such that $x \in U_1 \cap \dots \cap U_n$ and $U_1 \cup \dots \cup U_n$ is a neighborhood of x in X ;
- (b) for every $U, V \in \tau$, the collection of all $W \in \tau$ such that $W \subseteq U \cap V$ is a (resp. quasi-) net on $U \cap V$.

Yang: To be checked.

Definition 2. A $\mathbf{k}$ -*quasiaffinoid space* is a locally ringed space $(X, \mathcal{O}_X)$ which is isomorphic to an open subspace of a $\mathbf{k}$ -affinoid space.

Definition 3. A $\mathbf{k}$ -*analytic space* is a locally ringed space $(X, \mathcal{O}_X)$ such that every point $x \in X$ has an open neighborhood U such that $(U, \mathcal{O}_X|_U)$ is a $\mathbf{k}$ -quasiaffinoid space.

Remark 4. There are two different versions of the definition of $\mathbf{k}$ -analytic spaces in the literature: one given by Berkovich in

Definition 5. Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be $\mathbf{k}$ -analytic spaces. A *morphism* $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ of $\mathbf{k}$ -analytic spaces is a morphism of locally ringed spaces. The category of $\mathbf{k}$ -analytic spaces is the category whose objects are $\mathbf{k}$ -analytic spaces and whose morphisms are morphisms of $\mathbf{k}$ -analytic spaces. Yang: To be revised.

Theorem 6. Let $(X, \mathcal{O}_X)$, $(Y, \mathcal{O}_Y)$, and $(Z, \mathcal{O}_Z)$ be $\mathbf{k}$ -analytic spaces, and let $f : (X, \mathcal{O}_X) \rightarrow (Z, \mathcal{O}_Z)$ and $g : (Y, \mathcal{O}_Y) \rightarrow (Z, \mathcal{O}_Z)$ be morphisms of $\mathbf{k}$ -analytic spaces. Then the fibre product $(X, \mathcal{O}_X) \times_{(Z, \mathcal{O}_Z)} (Y, \mathcal{O}_Y)$ exists in the category of $\mathbf{k}$ -analytic spaces. Yang: To be revised.

Definition 7. Let $(X, \mathcal{O}_X)$ be a $\mathbf{k}$ -analytic space. We say that $(X, \mathcal{O}_X)$ is *good* if every point $x \in X$ has an open neighborhood U such that $(U, \mathcal{O}_X|_U)$ is a $\mathbf{k}$ -affinoid space.

2 The first properties

Definition 8. Let $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ be a morphism of $\mathbf{k}$ -analytic spaces. We say that f is *finite* if for every affinoid domain $V \subseteq Y$, the preimage $f^{-1}(V)$ is an affinoid domain in X and the induced morphism $\mathcal{O}_Y(V) \rightarrow \mathcal{O}_X(f^{-1}(V))$ is a finite morphism of $\mathbf{k}$ -affinoid algebras. **Yang:**

Theorem 9. Let $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ be a morphism of $\mathbf{k}$ -analytic spaces. Then there exists a factorization $f = g \circ h$ where $h : (X, \mathcal{O}_X) \rightarrow (Z, \mathcal{O}_Z)$ is a finite morphism and $g : (Z, \mathcal{O}_Z) \rightarrow (Y, \mathcal{O}_Y)$ is a morphism such that for every affinoid domain $W \subseteq Y$, the preimage $g^{-1}(W)$ is an affinoid domain in Z and the induced morphism $\mathcal{O}_Y(W) \rightarrow \mathcal{O}_Z(g^{-1}(W))$ is a flat morphism of $\mathbf{k}$ -affinoid algebras. **Yang:**

Definition 10. Let $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ be a morphism of $\mathbf{k}$ -analytic spaces. We say that f is *separated* if the diagonal morphism $\Delta_f : (X, \mathcal{O}_X) \rightarrow (X, \mathcal{O}_X) \times_{(Y, \mathcal{O}_Y)} (X, \mathcal{O}_X)$ is a closed immersion. **Yang:**

Definition 11. Let $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ be a morphism of $\mathbf{k}$ -analytic spaces. We say that f is *proper* if it is separated, compact, and without boundary. **Yang:**

Theorem 12. Let $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ be a proper morphism of $\mathbf{k}$ -analytic spaces. Then for every coherent $\mathcal{O}_X$ -module $\mathcal{F}$, the direct image $f_*(\mathcal{F})$ is a coherent $\mathcal{O}_Y$ -module. **Yang:**

Definition 13. Let $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ be a morphism of $\mathbf{k}$ -analytic spaces. We say that f is *smooth* (resp. *étale*) if for every $x \in X$, there exist open neighborhoods U of x in X and V of $f(x)$ in Y such that $f(U) \subseteq V$ and a commutative diagram **Yang:**

Appendix